Table Profiling

- Built once per table
- Reused across questions

T

src	Club	Founded	2022 Attendance (people)		2023 Attendance (k people)	
			League	Cup	League	Cup
r1	Tigers	1942	400	200	0.700	0.260
r2	Falcons	1955	800	100	1.000	0.120
r3	Lions	1968	1,200	180	1.500	0.200
r4	Bears	1974	1,000	150	1.180	0.170
...	...	...	...	...	...	...

Question-Independent Table Profiling

Table Profile

- Header hierarchy
- Types and units
- Representative values
- Stable cell addresses

1 Plan

What information must survive?

Backward Answerability Planning

Question q

Among clubs founded since 1950, which had the highest 2022–23 league-attendance growth rate? Give their 2023 attendance, including ties.

Table sketch S(T)

- Headers + shape
- Samples + rows
- Stable source IDs

Backward Planner

Answerability Signature

$A_0 = \langle D, L, R \rangle$

D	Scope	Clubs satisfying Founded >= 1950
L	Bindings	Club ↔ Attendance ↔ Years ↔ units
R	Required Distinctions	· Identity(Club) · Threshold(Founded >= 1950) · Exact(Attendance 2023) · Order↓ + Ties(GrowthRate = 100 × (A23 - A22) / A22)

Full-Source Grounding

Execute A_0 over Full Table T

Source check fails??

Correction

Signature Auditor

✓ Scope
✓ Binginds
✓ Expressions

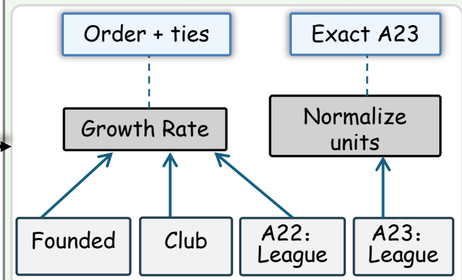
Accepted Signature A*

2 Compile

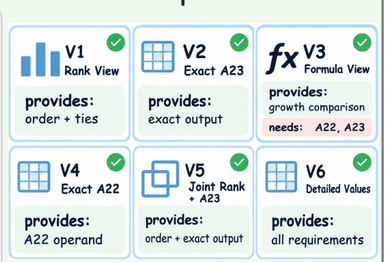
How should it be represented under budget?

Dependency-Aware Evidence Compilation

Answer Dependencies(Logical Plan)

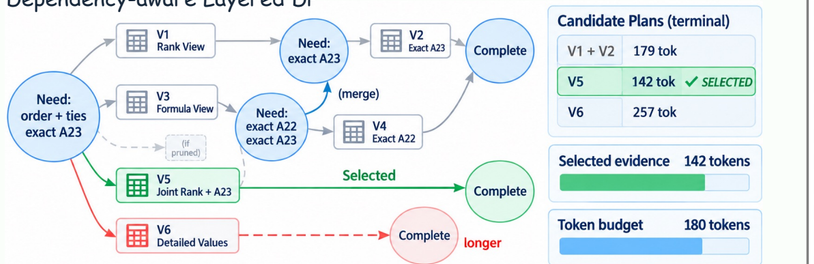


Candidate Representations



Physical Plan Optimization

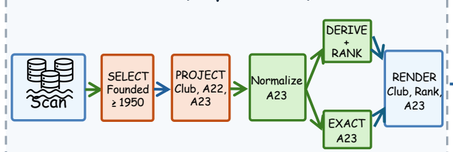
Dependency-aware Layered DP



$P^* = \text{argmin Cost}(\text{render}(P)), W^* = \text{render}(P^*)$

Selected Evidence Plan P*

(Physical Plan)



CompressTab witness W*

Club	Rank	A23
Falcons	#1	1,000
Lions	#1	1,500
Bears	#2	1,180
.....		

Scope: Founded >=1950
Rank scope: grounded population D*
Ties preserved; unit = people

3 Refine

Is the answer or evidence sufficiently supported?

Counterexample-Guided Refinement

q + W*

Answer Agent → "y_0 = Falcons--1000"

Evidence Auditor — "*Is y_0 supported*"?

Yes

No

Coverage|Bindings|Distinction|Reasoning|Output

Counterexample Reasoning

REASONING ERROR

"Falcons and Lions both have rank #1. y0 omits a tied winner.."

Answer Self-correction

"y_1 = Falcons-1000 and Lions-1500"

Missing evidence

"The answer depends on a unit note that is absent from A* and W*."

Counter-evidence Verification

Missing residual evidence A*

Expanded Signature

$A_1 = \text{Closure}(A^* \cup A^+)$

Recompilation

.....

COMMIT y_1
Repair verified

KEEP y_0
Critique invalid

UNRESOLVED
No certified revision

Final Answer